1 Isomorphisms

We have now seen a variety of examples of vector spaces. Some of these, however, are not essentially different. For instance, any singleton set $\{v\}$ can be given the structure of the zero vector space by defining $v+v=v$ and $\alpha v = v$ for all $\alpha \in F$. There is no real difference, then, between any two zero vector spaces $\{v\}$ and $\{w\}$; all we have done is label the only vector in the set by a different symbol. Algebraically, these are exactly the same. In general, two vector spaces are the same in this sense if

(i) there is a 1-1 correspondence (i.e., a bijective map) between the underlying sets, and

(ii) this 1-1 correspondence preserves vector addition and scalar multiplication; that is, it is linear.

Definition 1.1 (Isomorphism)**.** Let V and W be vector spaces over F. An *isomorphism from V to W* is a bijective (injective and surjective) linear map $T : V \to W$. If there exits an isomorphism from V to W, we say that V and W are *isomorphic* and write $V \cong W$.

Example 1.2. Define $T : \{v\} \to \{w\}$ by $Tv = w$. It is immediately verified that T is a bijection, and it is also linear since

$$\begin{aligned} T(\alpha v + v) &= Tv \quad (\text{since } \alpha v + v = v) \\ &= w \\ &= \alpha w + w \quad (\text{since } \alpha w + w = w) \\ &= \alpha Tv + Tv. \end{aligned}$$

Thus, T is an isomorphism from $\{v\}$ to $\{w\}$.

Exercise 1.3. Show that $F^n \cong M^{n\times 1}(F)$. One can similarly show that $F^n \cong M^{1\times n}(F)$. That is, ordered n-tuples can equivalently be represented as either row vectors or column vectors.

Exercise 1.4. Show that the subspace $W = \{(x, y, 0) \subseteq F^3\}$ is isomorphic to F^2.

Exercise 1.5. Show that $\mathscr{P}_n(F)$ (the vector space of all polynomials with coefficients in F of degree at most n) is isomorphic to F^{n+1}.

Exercise 1.6. Recall the "unusual" vector space V defined by $V = \mathbb{R}^+$ (the positive reals) with addition and multiplication defined by

$$x + y := xy$$
$$kx := x^k$$

Show that $V \cong \mathbb{R}$.

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2 Classification of finite-dimensional vector spaces

Proposition 2.1.

(a) For any vector space V, $id_V : V \to V$ is an isomorphism. Thus, $V \cong V$ for every V.

(b) If $T : V \to W$ is an isomorphism, then $T^{-1} : W \to V$ is also an isomorphism. Thus, $V \cong W$ implies $W \cong V$.

(c) If $T_1 : U \to V$ and $T_2 : V \to W$ are isomorphisms, then $T_2T_1 : U \to W$ is an isomorphism. Thus, $U \cong V$ and $V \cong W$ implies $U \cong W$.

Proposition 2.1 says that "isomorphic" is reflexive, symmetric, and transitive, and is therefore an equivalence relation on the collection of all vector spaces over F.

Exercise 2.2. Prove Proposition 2.1. We have previously proved that the identity map is linear and that the composition of linear maps is linear. You have most likely also proved in a previous course that the identity map is a bijection, the inverse of a bijection is a bijection, and that the composition of two bijections is a bijection (convince yourself of these facts if you need to). All that remains, then, is to prove that if $T : V \to W$ is a linear bijection, then $T^{-1} : W \to V$ is also linear.

Exercise 2.3. Exercise 2.1 says that "isomorphic" is an equivalence relation on the collection of vector spaces over F. Define the *isomorphism class*, $[V]$, of a vector space V to be the set of all vector spaces which are isomorphic to V: $[V] = \{W : W \cong V\}$. Show that

(a) Every vector space V over F belongs to some isomorphism class.

(b) Any two isomorphism classes are disjoint or equal.

Proposition 2.4. *Let V be a finite-dimensional vector space of dimension* $\dim V = n$ *and let* $B = (v_1, \ldots, v_n)$ *be an* ordered *basis of V. Define $T_B : \mathbb{R}^n \to V$ to be the linear combination map $T(x_1, \ldots, x_n) = x_1 v_1 + \cdots + x_n v_n$. Show that T_B is an isomorphism.*

Exercise 2.5. Prove Proposition 2.4.

Corollary 2.6. *Prove that if V and W are finite-dimensional vector spaces with* $\dim V = \dim W$*, then $V \cong W$.*

Exercise 2.7. Prove Corollary 2.6.

Classification of finite-dimensional vector spaces: We have just proved that there is a single isomorphism class of vector spaces over F for each non-negative integer. If $\dim V = 0$, then $V \cong \{0\}$. If $\dim V = n \geq 1$, then $V \cong F^n$.

Recall that for a map of finite sets $f : A \to B$ with $|A| = |B|$, injectivity *or* surjectivity implies bijectivity. The following result for linear maps between finite-dimensional vector spaces of the same dimension is analogous.

Proposition 2.8. *If V and W are finite-dimensional vector spaces of the same dimension and $T : V \to W$ a linear map, then if T is injective or surjective, it is an isomorphism.*

Exercise 2.9. Prove Proposition 2.8.

Exercise 2.10. Use Proposition 2.8 to prove that, given any polynomial q, there exists a polynomial p of the same degree such that $((x^2+5x+7)p)'' = q$.